ANANYO SEN

M.Sc. Data Science and Artificial Intelligence | Ramakrishna Mission Vivekananda Educational and Research Institute, Belur

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PROFESSIONAL SUMMARY

M.Sc. Data Science and Artificial Intelligence student with a B.Sc. (Honours) in Mathematics and academic project and research experience in machine learning, deep learning, reinforcement learning, computer vision, and distributed computing. Skilled in Python-based data analysis, exploratory data analysis, feature engineering, statistical methods, and model evaluation.

TECHNICAL SKILLS

Programming Languages: Python, R, JavaScript, C++, LaTeX
Data Science and Machine Learning: NumPy, Pandas, Scikit-learn, EDA, Data Preprocessing, Feature Engineering, Classification, Regression, Model Evaluation, Statistical Analysis, Hypothesis Testing, ANOVA
Deep Learning and Computer Vision: PyTorch, Torchvision, OpenCV, PyAV, OCR
Tools and Platforms: Ray, Flask, Gymnasium, Matplotlib, Seaborn, Streamlit, Git, GitHub
Operating Systems: Windows, Ubuntu/Linux, macOS

EDUCATION

Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI) 2025–2027
M.Sc. in Data Science and Artificial Intelligence | CGPA: 6.65 (through Semester 2) *Belur*
Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI) 2022–2025
B.Sc. (Honours) in Mathematics | CGPA: 6.91 *Belur*

TECHNICAL PROJECTS

Distributed Gaming Analytics System Using Ray *Feb 2026 – May 2026*
RKMVERI

- Developed a distributed gaming analytics platform using **Ray** to parallelize player-data analysis and predictive modelling; implemented a **Random Forest Regressor** and statistical analysis, and presented analytical results through a **Flask** web application.
- *Technologies:* Python, Ray, Flask, Random Forest Regression, Predictive Analytics, Statistical Analysis.

Event-Driven Multi-Modal Sports Video Understanding *Feb 2026 – Apr 2026*
RKMVERI

- Investigated deep learning methods for sports video understanding, including temporal event detection, tactical interaction modelling, match-intensity estimation, and automated commentary generation.
- *Technologies:* Python, PyTorch, Torchvision, PyAV, SoccerNet; Computer Vision, Video Analysis.

Deep Q-Networks for Partially Observable Treasure Hunt *Oct 2025 – Dec 2025*
RKMVERI

- Developed a **Deep Q-Network (DQN)** agent for a partially observable environment with limited visibility, dynamic obstacles, and sparse rewards; designed adaptive difficulty levels to evaluate navigation under uncertainty.
- *Technologies:* Python, Gymnasium, Deep Q-Learning, Reinforcement Learning, Markov Decision Processes.

Prediction of Video Game Sales *Sep 2025 – Nov 2025*
RKMVERI

- Performed exploratory data analysis, data preprocessing, feature engineering, and predictive modelling to estimate global video game sales; compared machine learning models and evaluated performance for revenue forecasting.
- *Technologies:* Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn; Decision Tree, Logistic Regression, Random Forest, Gradient Boosting.

RESEARCH EXPERIENCE

Statistical Methods and Literature Review for Machine Learning in Mental Health *May 2026 – Jul 2026*
RKMVERI

- Applied statistical research methods including hypothesis testing, Analysis of Variance (ANOVA), statistical power analysis, sample-size estimation, and non-parametric tests; reviewed machine learning and psychometric validation studies, including SERATS, SERAPTS, and FERATS, and presented findings in academic seminars.

Summer Intern – FLAME School of Computing and Data Sciences *May 2026 – Sep 2026*
FLAME University, Pune | Supervisor: Dr. Chiranjoy Chattopadhyay | View Internship Certificate

- Completed a literature-based project, *Multimodal Understanding of Comics and Visual Narratives*, focused on multimodal learning and visual document understanding.
- Reviewed comic-panel extraction, reading-order prediction, stylized image processing, OCR, multimodal emotion recognition, cross-panel reasoning, narrative understanding, and the ComicsPAP vision-language benchmark; synthesized challenges into a framework for future research.

ACADEMIC RESEARCH PROJECT

Properties of Bessel Functions *Jan 2025 – Apr 2025*
Ramakrishna Mission Vidyamandira

- Studied recurrence relations, generating functions, orthogonality, and Fourier–Bessel expansions, including applications to Laplace's equation in cylindrical coordinates.

RELEVANT COURSEWORK

Machine Learning, Artificial Intelligence, Deep Learning, Probability and Statistics, Linear Algebra, Time Series Analysis, Graph Theory, Distributed Computing, Data Structures and Algorithms

ACHIEVEMENTS

- Qualified, CUET-PG 2025 (Mathematics).
- JBNSTS Junior Scholarship, Government of West Bengal (2020).
- First Place, Puzzle Solving, Enigma Quest, Perceptron'25, RKMVERI.
- Second Place, Poster Presentation on "Real Analysis and Its Applications", Ramakrishna Mission Vidyamandira.

CERTIFICATIONS

- NPTEL – Programming in Modern C++ (2025).
- NPTEL – Generative AI for Computer Vision (2026).
- NPTEL – Introduction to Large Language Models (2026).